

- 2** For $f(x, y) = e^{xy} \sin(x + y)$, find f_x using the product rule.

- 4** Find the directional derivative of $f(x, y) = x^2 + y^2$ at $(1, 1)$ in the direction of $\mathbf{v} = \langle 3, 4 \rangle$.

- 6** Find the equation of the tangent plane to $z = xy$ at the point $(2, 3, 6)$.

- 8** Verify that the critical point found above is a minimum, using the second derivative test with $D = f_{xx}f_{yy} - (f_{xy})^2$.

- 9** Let $z = x^2y$, where $x = t^2$ and $y = t^3$. Use the multivariable chain rule $\frac{dz}{dt} = \frac{\partial z}{\partial x} \frac{dx}{dt} + \frac{\partial z}{\partial y} \frac{dy}{dt}$ to find $\frac{dz}{dt}$ at $t = 1$.
- 10** Let $w = x^2 + y^2 + z^2$, where $x = \cos t$, $y = \sin t$, $z = t$. Find $\frac{dw}{dt}$ using the chain rule.

Maximum rate of increase

- 11 Find the direction in which $f(x, y) = x^2 + xy$ increases most rapidly at the point $(1, 2)$, and state the maximum rate of increase.
- 12 Find the direction of steepest descent for $f(x, y) = x^2 + y^2$ at the point $(3, 4)$.
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Implicit partial differentiation

- 13 For the surface $x^2 + y^2 + z^2 = 9$, find $\frac{\partial z}{\partial x}$ by implicit differentiation (treating y as constant).